should not result in worse than minor effects.
13.2 Numerical values for various levels of probabilities are laid down in section 3 of annex 3 of this Code. In areas where there is no data from craft to determine the level of probabilities of failure other sources can be used such as:
.1 workshop test, or

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SOURCE: MSC.36(63).pdf
section 3 of annex 3 of this Code. In areas where there is no data from craft to determine the level of probabilities of failure other sources can be used such as:
.1 workshop test, or
.2 history of reliability used in other areas under similar operating conditions, or
.3 mathematical model if applicable.
14 Documentation
14.1 It is helpful to perform FMEA on worksheet(s) as shown in appendix 2.
14.2 The worksheets(s) should be organized to first display the highest system level and then proceed down through decreasing system levels.
15 Test programme
15.1 An FMEA test programme should be drawn up to prove the conclusions of FMEA. It is recommended that the test programme should include all systems or system elements whose failure would lead to:
.1 major or more severe effects;
.2 restricted operations; and
.3 any other corrective action.
For equipment where failure cannot be easily simulated on the craft, the results of other tests can be used to determine the effects and influences on the systems and craft.
15.2 The trials should also include investigations into:
.1 the layout of control stations with particular regard to the

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SOURCE: MSC.36(63).pdf
results of other tests can be used to determine the effects and influences on the systems and craft.
15.2 The trials should also include investigations into:
.1 the layout of control stations with particular regard to the relative positioning of switches and other control devices to ensure a low potential for inadvertent and incorrect crew action, particularly during emergencies, and the provision of interlocks to prevent inadvertent operation for important system operation;
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.2 the existence and quality of the craft's operational documentation with particular regard to the pre-voyage checklists. It is

essential that these checks account for any unrevealed failure modes identified in the failure analysis; and
.3 the effects of the main failure modes as prescribed in the theoretical analysis.
15.3 The FMEA tests on board should be conducted in conjunction with provisions specified in 5.3, 16.4 and 17.4 of this Code, before the craft enters into service.
16 FMEA Report

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SOURCE: MSC.36(63).pdf
theoretical analysis.
15.3 The FMEA tests on board should be conducted in conjunction with provisions specified in 5.3, 16.4 and 17.4 of this Code, before the craft enters into service.
16 FMEA Report
The FMEA report should be a self-contained document with a full description of the craft, its systems and their functions and the proposed operation and environmental conditions for the failure modes, causes and effects to be understood without any need to refer to other plans and documents not in the report. The analysis assumptions and system block diagrams should be included, where appropriate. The report should contain a summary of conclusions and recommendations for each of the systems analysed in the system failure analysis and the equipment failure analysis. It should also list all probable failures and their probability of failure, where applicable, the corrective actions or operational restrictions for each system in each of the operational modes under analysis. The report should contain the test programme, reference any other test reports and the FMEA trials.
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SOURCE: MSC.36(63).pdf
in each of the operational modes under analysis. The report should contain the test programme, reference any other test reports and the FMEA trials.
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APPENDIX 1
EXAMPLE OF A SYSTEM BLOCK DIAGRAM
Steering Control System
EP "-.0 CcrftI EP SI-.o~ '-_21S2l ~_I(sl)
ES
EP
~<S3)
ES Selector

EP ES
SI_~
es "'-'0,"
HP ES
!AS
where:
EP - electric power
HP - hydraulic power
ES - electric signal
MS - mechanical signal
W/1689n/pp/EWP
Date _____ ___
Analyst _____ ___
~SVtfem
EI? SI~ EP Indcattcn (11) ~(81)
ES
I
I
ES I
I
0Ia-Qe0i. EP. IndJcallon ConrOo \II2) Controller (12)
ES
EP
Indcatng
SenIo (13)
ES
MS
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TABLE 1
EXAMPLE OF A SET OF FAILURE MODES
1 Structural failure 18
(rupture)
2 Physical binding or jamming 19
3 Vibration 20
4 Fails to remain (in 21
position)
5 Fails to open 22

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TABLE 1
EXAMPLE OF A SET OF FAILURE MODES
1 Structural failure 18
(rupture)
2 Physical binding or jamming 19

3 Vibration 20
4 Fails to remain (in 21
position)
5 Fails to open 22
6 Fails to close 23
7 Fails open 24
8 Fails closed 25
9 Internal leakage 26
10 External leakage 27
11 Fails out of tolerance (high) 28
12 Fails out of tolerance (low) 29
13 Inadvertent operation 30
14 Intermittent operation 31
15 Erratic operation 32
16 Erroneous indication 33
-
17 Restricted flow
Refer to the following IEC Publication :'812 IEC 1985
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False actuation
Fails to stop
Fail~ to start
Fails to switch